Project Planning & Management Document

NutriVision

1. Project Overview

NutriVision is an AI-powered dietary health assistant designed to analyze user-entered meal descriptions and generate personalized nutritional insights and healthier recipe alternatives. The application integrates Artificial Intelligence, cloud technologies, and modern UI/UX principles to provide real-time dietary tracking and synchronized meal logs across multiple devices.

2. Project Objectives

The project aims to:

- Provide accurate nutritional analysis of meals.
- Generate AI-based healthier alternatives tailored to dietary restrictions.
- Enable real-time synchronization using Firebase Cloud Firestore.
- Ensure a responsive and intuitive user interface.
- Validate system performance through structured testing.

3. Project Scope

3.1 In Scope

- Meal description analysis
- AI-generated recipe suggestions
- Real-time meal logging
- Cross-device synchronization
- Unit, widget, and integration testing

3.2 Out of Scope

- Medical diagnosis or treatment
- Wearable device integration
- Payment or subscription systems

4. Project Management Methodology

The project follows the **Agile methodology using Scrum framework**.

Scrum Structure:

- Sprint Duration: 2 Weeks
- Sprint Planning
- Incremental Development
- Sprint Review & Retrospective

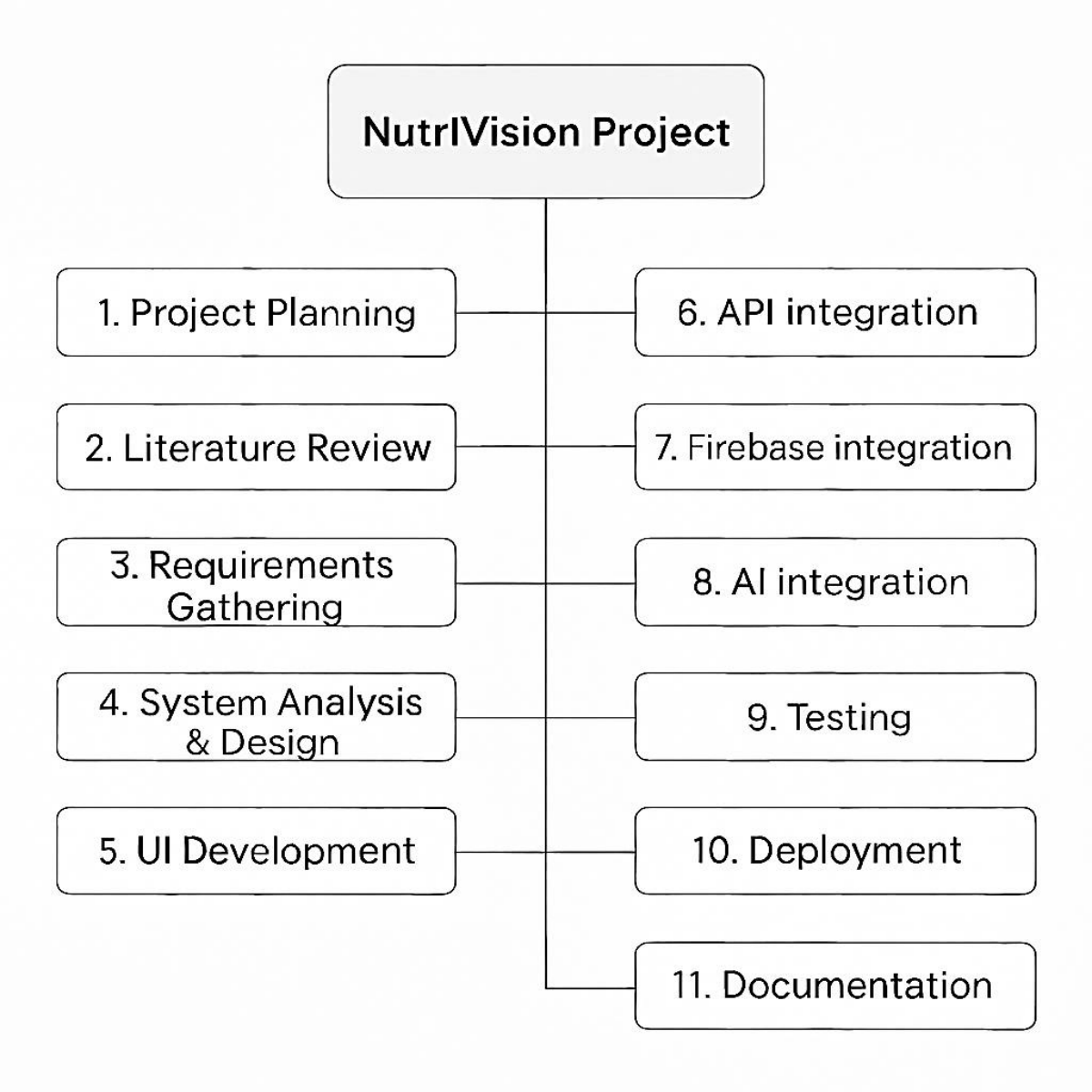
Justification:

Agile allows flexibility in refining AI prompts, improving UI/UX design, and managing API integrations effectively.

5. Work Breakdown Structure (WBS)

1. Project Planning
2. Literature Review
3. Requirements Gathering
4. System Analysis & Design
5. UI Development
6. API Integration
7. Firebase Integration
8. AI Integration
9. Testing
10. Deployment
11. Documentation

5.1 WBS diagram showing the hierarchy of the project phases



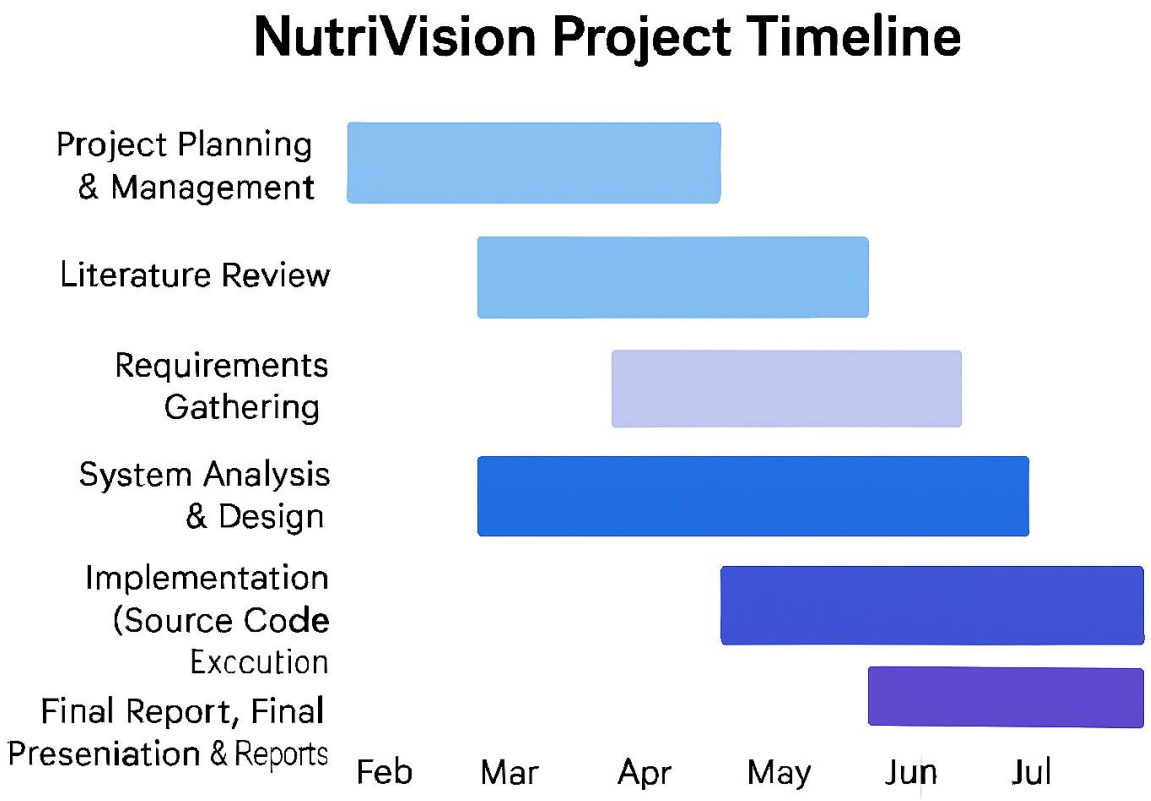
6. Project Timeline

NutriVision Timeline (Aligned with Official Deadlines)

Phase	Activity	Start Date	End Date	Official Submission Date	Deliverable
Phase 1	Project Planning & Management	Feb 1, 2026	Feb 20, 2026	**Feb 20, 2026**	Project Management Plan
Phase 2	Literature Review	Mar 1, 2026	Apr 20, 2026	**Apr 20, 2026**	Literature Review Report

Phase 3	Requirements Gathering	Mar 25, 2026	Apr 20, 2026	**Apr 20, 2026**	Software Requirements Specification (SRS)
Phase 4	System Analysis & Design	Apr 21, 2026	May 1, 2026	**May 1, 2026**	System Design Document (UML, Architecture, ERD)
Phase 5	Implementation (Source Code & Execution)	May 2, 2026	Jul 10, 2026	**Jul 10, 2026**	Functional Application + Source Code
Phase 6	Testing, Final Presentation & Reports	Jul 11, 2026	Jul 17, 2026	**Jul 17, 2026**	Final Report + Presentation + Testing Documentation

6.1 Gantt chart for the project milestones



7. Risk Management Plan

Risk	Probability	Impact	Mitigation Strategy
Inaccurate AI outputs	Medium	High	Prompt refinement and validation logic
API downtime	Medium	High	Implement error handling & fallback responses
Firebase sync delay	Low	Medium	Optimize queries and indexing
Performance issues	Medium	Medium	Code optimization & profiling

8. Quality Management Plan

Quality assurance will include:

- Unit Testing for business logic
- Widget Testing for UI components
- Integration Testing for API & Firebase
- User Acceptance Testing

Quality Metrics:

- API response time < 2 seconds
- Real-time sync delay < 1 second
- Stable cross-device synchronization
- High test coverage

9.Resource Management

Development Tools

- Flutter & Dart
- Firebase Cloud Firestore
- Dio Package
- AI APIs

Hardware Resources

- Android Emulator
- Physical Android Device

10.Deliverables

- Fully functional mobile application
- Source code repository
- Architecture & UML diagrams
- Testing documentation
- Final report
- Presentation slides
- APK build

11.Success Criteria

The project will be considered successful if:

- Nutritional analysis functions accurately.
- AI generates relevant dietary suggestions.
- Real-time synchronization works correctly.
- The application is stable and user-friendly.
- All submission deadlines are met.

12.Work Distribution Strategy

Each phase will be split into 5 parallel tasks, and each member will rotate across phases to ensure balanced contribution and shared ownership. Here's how it works:

Phase	Tasks	Assigned Members
1. Project Planning	Timeline, WBS, Risk Plan, KPIs, Roles	Ahmed, Jessica, Nouran, Omar A., Omar S.
2. Literature Review	Source collection, summarization, citation formatting, relevance analysis, report writing	Ahmed, Jessica, Nouran, Omar A., Omar S.
3. Requirements Gathering	Functional reqs, non-functional reqs, use cases, SRS formatting, review	Ahmed, Jessica, Nouran, Omar A., Omar S.
4. System Analysis & Design	UML diagrams, architecture, ERD, design rationale, documentation	Ahmed, Jessica, Nouran, Omar A., Omar S.
5. UI Development	Wireframes, layout coding, responsiveness, accessibility, testing	Ahmed, Jessica, Nouran, Omar A., Omar S.
6. API Integration	API selection, Dio setup, interceptors, error handling, testing	Ahmed, Jessica, Nouran, Omar A., Omar S.
7. Firebase Integration	Firestore setup, sync logic, multi-device testing, security rules	Ahmed, Jessica, Nouran, Omar A., Omar S.
8. AI Integration	Prompt design, recipe generation logic, fallback handling, testing	Ahmed, Jessica, Nouran, Omar A., Omar S.
9. Testing	Unit tests, widget tests, integration tests, test reports, bug fixing	Ahmed, Jessica, Nouran, Omar A., Omar S.
10. Deployment	Build optimization, release packaging, store submission, feedback loop	Ahmed, Jessica, Nouran, Omar A., Omar S.
11. Documentation	Final report, presentation slides, diagrams, formatting, proofreading	Ahmed, Jessica, Nouran, Omar A., Omar S.